EXP.NO:	6	SHARED MEMORY IPC
DATE:		

AIM

To implement Inter-Process Communication (IPC) using Shared Memory in Linux using C, demonstrating how two processes can communicate by sharing a common memory segment.

PROCEDURE

1. Writer process creates a shared memory segment using shmget().
2. It attaches the memory using shmat() and writes data into it.
3. Reader process attaches to the same shared memory and reads the data.
4. Both processes detach and delete the shared memory using shmdt() and shmctl().

PROGRAM – Writer (writer.c)

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/shm.h>
#include <sys/ipc.h>

#define KEY 1234
#define SIZE 1024

int main() {
    int shmid = shmget(KEY, SIZE, IPC_CREAT | 0666);
    if (shmid < 0) { perror("shmget failed"); exit(1); }

    char *shm = (char *) shmat(shmid, NULL, 0);
    if (shm == (char *) -1) { perror("shmat failed"); exit(1); }

    printf("Writer: Enter message: ");
    fgets(shm, SIZE, stdin);
    printf("Writer: Message written to shared memory.\n");

    shmdt(shm);
    return 0;
}
```

PROGRAM – Reader (reader.c)

```
#include <stdio.h>
#include <stdlib.h>
#include <sys/shm.h>
#include <sys/ipc.h>

#define KEY 1234
#define SIZE 1024

int main() {
    int shmid = shmget(KEY, SIZE, 0666);
    if (shmid < 0) { perror("shmget failed"); exit(1); }

    char *shm = (char *) shmat(shmid, NULL, 0);
    if (shm == (char *) -1) { perror("shmat failed"); exit(1); }

    printf("Reader: Message from shared memory: %s", shm);

    shmdt(shm);
    shmctl(shmid, IPC_RMID, NULL);
    printf("Reader: Shared memory released.\n");
    return 0;
}
```

OUTPUT

```
--- Terminal 1: Writer ---
$ gcc writer.c -o writer
$ ./writer
Writer: Enter message: Hello from shared memory!
Writer: Message written to shared memory.

--- Terminal 2: Reader ---
$ gcc reader.c -o reader
$ ./reader
Reader: Message from shared memory: Hello from shared memory!
Reader: Shared memory released.
```

RESULT

Inter-Process Communication using Shared Memory was successfully implemented. The writer process wrote data into shared memory and the reader process retrieved it correctly, demonstrating IPC between two processes.